Fiscal Year:	FY 2019	**Task Last Updated:**	FY 01/25/2019
PI Name:	Weislogel, Mark M. Ph.D.		
Project Title:	Applied Capillary Fluidics Research For Spacecraft		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Portland State University		
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Zip Code:	97207-0751	**Congressional District:**	3
Comments:	Professional affiliations for Dr. Weislogel (updated 1/2/2025): IRPI, LLC 27501 SW 95th Ave, Suite 930 Wilsonville, OR 97070-2905 and Portland State University MCECS Research, P.O. Box 751 Portland, OR 97207-0751		
Project Type:	Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	01/19/2018	**End Date:**	08/20/2021
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/20/2021 per NSSC information (Ed., 8/17/21) NOTE: End date changed to 4/10/2021 per NSSC information (Ed., 9/3/20) NOTE: End date changed to 4/10/2020 per NSSC information (Ed., 10/21/19) NOTE: End date changed to 1/18/2020 per NSSC information (Ed., 2/27/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K0436		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	To meet the challenges posed by NASA's plans to further explore outer space, a large body of applied science and engineering investigations must be completed to assure the development, construction, and demonstration of the highly reliable performance of myriad advanced life support systems and unit operations. The advances must occur across all subjects within the Physical Sciences discipline, but perhaps most pointedly within the Fluid Physics sub-discipline. From liquid fuels, to propellants, to cryogens, to coolants, to water processes and management, to plant and animal habitats, and others, the management of liquids without the benefit of constant gravitational separation poses significant challenges to the designers of such systems. In this proposed work we seek to apply the results of recent low-g capillary fluidics research, performed in part aboard the International Space Station (ISS) and Portland State University Dryden Drop Tower, to deliver solutions to NASA for a variety of pressing needs. These include: Passive biofluids management, Water delivery and management for plants in space, and a variety of low-g biofluids, surface tension containment, and educational/outreach demonstrations and activities.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The majority of the effort of this research centers on the development of plant watering systems for spacecraft, which have been troubling since the beginning of space flight. We are supporting modern design activities and down-selection of method(s) for water delivery to a variety of plant habitats considering hydroponics, and artificial soil for single- and multi-plant arrays. Positive research returns could mean a new era of plant growth capabilities in space. The approach takes into account low-g and partial g environments such as lunar and Martian g environments.
Task Progress:	We are delivering quantifiable engineering solutions to a variety of fluids problems that can be addressed via modern capillary fluidics solutions. As most beneficial, and in connection with NASA Glenn Research Center (GRC) leadership, we provided and will continue to provide direct consulting support, low-g drop tower test development and support, data reduction and analysis, prototype development, and article co-authorship as merited. We identified target application problems for the first year of activity. For brevity we provide progress below only for water delivery and management for plants in space. We supported modern design activities and down-selection of method(s) for water delivery to a variety of plant habitats considering hydroponics, and artificial soil for single- and multi-plant arrays. Terrestrial and drop tower tests were conducted for aspects of the designs requiring increased TRL (technology readiness level). Support to NASA for the implementation of novel designs were provided for the short delivery schedule technology development flight demonstrations. Two visits with NASA GRC personnel to NASA Kennedy Space Center (KSC) were conducted to make beneficial connection with Plant Biologists critical to the accurate representation of the hardware for ISS development. Twice weekly meetings were also supported with NASA GRC/contractor and NASA GRC/KSC teams. Science Requirements Documents (SRDs) were completed for the first 2 of 6 devices with various safety reviews supported leading to acceptance of the design before 2018 year end. The remaining 4 units SRDs remain largely complete requiring only final specifications of objectives pending long term project support plans not fully decided by NASA at present. Patent protection is being sought for the hydroponic solution which is currently envisioned for highly parallelized deployment in larger plant habitats.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Abstracts for Journals and Proceedings	Ungar E, Weislogel M. "Surface Tension Containment Experiment (STCE)- Increasing Science Throughput on ISS." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. ID-259. , Nov-2018
Abstracts for Journals and Proceedings	Mungin R, Weislogel D, Ringle M, Callahan M, Chen Y, Sargusingh M. "Evaporation in microgravity: recent results from the ISS CSELS experiment." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. ID-271. , Nov-2018
Abstracts for Journals and Proceedings	Weislogel M, Graf J, Hinojosa C, McQuillen J, Sargusingh M, Venkateswaran, K, Wetzel J, Wheeler R. "The done, the doable, and the daunting challenges of plumbing aboard spacecraft: Advanced water management for life support." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. ID-232. , Nov-2018